AI Procurement Request Copilot

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1. Establish the problem

Employees request software without a consistent view of existing tools, budget, vendor risk or approvals. Procurement needs a traceable next step, not an automated purchase. This product turns a request into a source-backed review packet while leaving sensitive decisions with humans.

Success: material gaps and conflicting evidence are visible; financial thresholds are applied by code; each request has a clear recommendation, evidence, approvals, missing information, risk flags and next step. All course data is synthetic.

2. Working vertical slice

The editable request UI gathers budget, software catalog, vendor registry/API/purchase history and policy evidence. A fifth deterministic tool applies amount thresholds, budget checks, security/privacy/legal rules and review freshness. Evidence is cached within a request, referenced in the output and visible in an expandable panel. A download action creates a human review packet; it does not approve or send it.

One-command local start: **./start.sh**. It creates a virtual environment, installs packages, starts the mock vendor API (8001) and Streamlit UI (8501). Offline mode works without credentials. Live mode uses a local Gemini API key (GEMINI_API_KEY, model set by MODEL_NAME, default gemini-2.5-flash). No secrets are included.

3. Architecture experiment

A - Single agent	B - Staged / two agents
UI -> Procurement Agent -> code controls -> human	UI -> Procurement Analyst -> Policy/Risk Reviewer -> code controls -> human
One attempted LLM call in live mode	Two attempted LLM calls in live mode
Shared evidence pack and five read-only tools	Same evidence pack; no tool re-fetch

The diagram and workflow assumptions are in docs/architecture.svg. AI interprets need and recommends. Code owns thresholds and controls. Humans own approvals and exceptions.

Evaluation and ship decision

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4. Measured evidence

The same ten disclosed requests ran three times per architecture against the real local mock API. The six course public cases are included. Additional cases cover PII/cross-region, expired and conflicting vendor records, an existing tracker, and a low-value service request.

Offline measurement	A	B
Minimum behavioral checks	30 / 30	30 / 30
Source-reference / code-control / human checks	30 / 30 each	30 / 30 each
Tools per run	5	5
Actual LLM calls	0	0
Median local latency (ms)	16.72	13.79
p95 local latency (ms)	23.72	19.68

These are identical-path offline rule-preview measurements, not an AI architecture comparison. Equal scores do not establish A/B semantic equivalence. Source-reference checks do not independently grade the truth of model reasoning. Raw decisions, per-run CSV and reproducible runner are included. Local timing is environment-specific.

40 tests pass: starter integrity plus threshold boundaries, invalid amounts, API failure, missing data, injection, cache reuse, model failure, code controls that a model cannot remove, and unknown data classes. The course public minimum checks pass 6/6 in both offline architectures.

5. Reliability and assumptions

Policy snapshot date is 2026-09-30, not the computer clock. Review validity is at most 365 days; future, expired, missing or conflicting vendor evidence cannot imply approval. Cost is annual USD. Catalog seats are licensed capacity, not free seats. Overlap triggers review, not rejection. Request and vendor text is untrusted. Unknown data classes escalate. Missing/invalid AI output in live mode routes to manual review. No purchase, budget or legal-acceptance tools exist.

6. Provisional ship decision and remaining gates

Prefer A provisionally for lower orchestration and fewer model calls, but do not ship from offline evidence alone. B should earn its extra call through measurable safety or semantic benefit. The decision memo is under 500 words. Before final submission: provide an authorized model key, run both architectures on the same set, inspect semantic grounding and actions, complete the final code review, and record the Loom demo. A source-level Claude review identified issues; revised code is awaiting re-review.

Sources: supplied Assessment 3 brief; original starter pack policy/data/public evaluations. Full provenance and setup are in README.md. GitHub public publication and form submission remain held for the student's explicit review and go.